

AIFS-CRPS: ensemble forecasting using a model trained with a loss function based on the continuous ranked probability score

Source Information

"AIFS-CRPS: ensemble forecasting using a model trained with a loss function based on the continuous ranked probability score", Simon Lang et al., ECMWF, *npj Artificial Intelligence* 2:18 (2026), doi:10.1038/s44387-026-00073-7. 12 pages.

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Research Objective

Whether a single trained model, optimised directly against a proper scoring rule, can replace a physics-based ensemble at operational medium range and remain competitive at subseasonal range — and do so without the per-member training cost of diffusion-based ensemble models (s44387-026-00073-7, p.1).

Problem

Ensembles are the only operational route to forecast uncertainty, and the IFS ensemble is expensive: it requires perturbed data assimilations and singular vector perturbations. Meanwhile first-generation machine-learned weather models were deterministic, MSE-trained single forecasts whose ensemble members are separate models with too little spread. Diffusion-based machine-learned ensembles were the existing alternative, but they typically require multiple denoising steps per forecast step. The paper takes a different route: model the uncertainty directly in the loss (s44387-026-00073-7, p.2).

Gap Addressed in paper

The specific gap is scoring-rule degeneracy. The fair CRPS (fCRPS) corrects CRPS for finite ensemble size, but it degenerates: if all members except one equal the verifying observation, the remaining member is unconstrained and can take any value without changing the score. Machine-learned models trained in float16 or lower make this more likely, and raising ensemble size to mitigate it scales compute linearly (s44387-026-00073-7, p.9).

The fix is the almost fair CRPS, a convex combination $\text{afCRPS}_{\alpha} = \alpha * \text{fCRPS} + (1 - \alpha) * \text{CRPS}$, with α a hyperparameter in $(0, 1]$, $\alpha = 1$ recovering fCRPS, and training run at $\alpha = 0.95$. The authors argue the correspondence to an unperturbed control member is natural here, because training is inherently probabilistic (s44387-026-00073-7, p.9).

Findings and conclusion

A 50-member 15-day AIFS-CRPS ensemble beats the 9 km 50-member IFS ensemble for most upper-air variables, with improvements in the 5-20% range; scores at 100 hPa and above can degrade (s44387-026-00073-7, p.2).

The headline is not accuracy but *calibration behaviour*. Unlike AIFS with an MSE loss, which loses small-scale detail with lead time, AIFS-CRPS members maintain variability close to the training distribution throughout the forecast range, with no damping of smaller scales visible (s44387-026-00073-7, p.2). That is the direct answer to the blurring problem the first AIFS paper accepted.

At subseasonal range — up to 46 days, evaluated on 8-member reforecasts over 2018-2022 — AIFS-CRPS is competitive with or better than operational IFS reforecasts despite being trained only on forecasts up to 72 hours, including improved MJO predictions. The ECMWF ensemble spread would normally be inflated for reliability, but AIFS-CRPS ensemble-mean RMSE is substantially lower, so such inflation is likely unnecessary (s44387-026-00073-7, p.6).

AIFS-ENS, the operational real-time system, is based on the AIFS-CRPS N320 configuration described here (s44387-026-00073-7, p.7).

Limitations or Weakness

Calibration is not uniformly right. AIFS-CRPS tends to be over-dispersive in the extra-tropics, with ensemble spread larger than ensemble-mean RMSE, most visibly for 500 hPa geopotential,

and the spread-error correspondence is worse than for the IFS ensemble. In the tropics the sign reverses: spread is notably too small (s44387-026-00073-7, p.3).

The authors name the sharpest limitation themselves: CRPS can have limited sensitivity to the tail properties of distributions, so assessment of extreme-event skill remains an open next step (s44387-026-00073-7, p.6). This sits awkwardly beside the field's main argument for probabilistic models.

Stratospheric skill is reduced, which the authors attribute to the afCRPS objective's high sensitivity to vertical loss scaling; revised scalings are under consideration. Score degradation at 100 hPa in the N320 configuration is likewise linked to pressure scaling differences.

The N320 and O96 configurations are not cleanly comparable: N320 trains with two ensemble members instead of four and does not yet use the minimum pressure scaling factor, and the authors flag these as choices to be revised.

The MJO result is explicitly hedged as a single example whose propagation characteristics may not generalise.

Finally, initial conditions come from the operational IFS ensemble, so AIFS-CRPS inherits IFS's initial-condition uncertainty representation — an acknowledged simplification for a paper whose thesis is replacing IFS.

Implication or suggestions on future research

Include perturbed initial conditions during training rather than only at inference. Test extreme-event skill with a score that is sensitive to distribution tails. Revise the vertical loss scaling to recover the stratosphere. Explore higher-resolution states, revised initial-condition perturbations, and more forecast parameters (s44387-026-00073-7, p.7).

How your search can fill gap

The cleanest open question is the one the paper's own design cannot answer: how much of the ensemble gain comes from the *loss function* rather than from the extra compute a probabilistic objective permits. A fair test is a CRPS-shaped but non-proper baseline — for example an MSE-trained model evaluated as an ensemble, or CRPS with alpha fixed low enough to be improper — trained on identical data at identical resolution with an identical member budget.

Second, the over-dispersion in the extra-tropics and under-dispersion in the tropics is a spatially structured miscalibration, not noise. Post-hoc spread calibration applied separately per region and variable would test whether the raw ranking is sound and only the spread needs rescaling.

Third, because CRPS is tail-insensitive by construction, pairing this line of work with a tail-sensitive score is the most direct route to a defensible claim about extremes.